

CHMAS: A Coupled Hierarchical Framework for Multi-Agent Reinforcement Learning

Dongming Wang, Jie Xu, Yanyu Zhang, and Wei Ren

Abstract—Multi-agent reinforcement learning (MARL) systems face fundamental challenges in balancing global coordination with local execution across different temporal scales. This paper introduces the Coupled Hierarchical Multi-Agent System (CHMAS), a novel framework that decomposes multi-agent decision-making into centralized strategic planning and distributed tactical execution with bidirectional information flow. The strategic layer integrates all agents’ states with an exclusive global environmental state to generate guidance actions every T timesteps, while tactical agents execute distributed policies augmented by strategic guidance and local neighborhood observations. Unlike existing hierarchical approaches with unidirectional control, CHMAS establishes a feedback mechanism where accumulated tactical rewards influence strategic objectives through a coupling coefficient λ , ensuring strategic plans remain grounded in tactical feasibility. To address the non-stationarity inherent in hierarchical learning, we propose an asynchronous update protocol where strategic parameters update every N_f tactical episodes, allowing tactical policies to converge to quasi-stationary points between strategic changes. We present both a general bi-level formulation capturing full system dynamics and a tractable additive approximation enabling rigorous analysis. Theoretical analysis proves that this asynchronous scheme achieves $\mathcal{O}(\log K/\sqrt{K})$ convergence for the strategic layer after K strategic updates under standard assumptions. Experimental validation in a multi-agent foraging domain demonstrates successful learning of spatially partitioned exploration strategies, with both layers converging stably despite hierarchical coupling.

I. INTRODUCTION

Multi-agent reinforcement learning (MARL) has emerged as a fundamental paradigm for autonomous coordination in complex environments, with applications spanning robotics [1], traffic management [2], and resource allocation [3]. Despite significant progress, real-world deployment faces a critical challenge: coordinating distributed agents across multiple temporal and spatial scales while maintaining computational tractability.

The curse of dimensionality renders purely centralized approaches intractable. For N agents with state spaces $\mathcal{S}_i$ and action spaces $\mathcal{A}_i$, the joint space grows as $\mathcal{O}(\prod_{i=1}^N |\mathcal{S}_i| \cdot |\mathcal{A}_i|)$ [4]. While distributed methods address scalability through local decision-making [5], they sacrifice coordination quality for long-horizon objectives. Traditional approaches like QMIX [6] and MADDPG [7] leverage centralized training with decentralized execution, but assume all agents access relevant global information during training, which often fails

in practice due to partial observability and communication constraints [8].

Furthermore, multi-agent systems inherently operate across multiple temporal scales. Strategic decisions (formation control, task allocation) require global information including environmental states inaccessible to individual agents, operating on extended timescales. Tactical responses (collision avoidance, local adaptation) demand rapid reactions based on immediate observations. While hierarchical reinforcement learning offers a natural framework for multi-scale decision-making [9], [10], existing hierarchical MARL approaches [11], [12] employ unidirectional control where high-level policies dictate low-level actions, missing opportunities for tactical feedback to inform strategic planning.

The interaction between hierarchical levels creates complex learning dynamics. Strategic changes fundamentally alter the environment perceived by tactical agents, violating stationarity assumptions [8]. Conversely, evolving tactical policies change execution characteristics that strategic planners rely upon. Previous hierarchical MARL work [13], [14] addresses this through reward engineering or option learning but requires domain expertise and often fails to generalize.

This paper introduces CHMAS, which addresses these limitations through bidirectional coupling between strategic and tactical layers at different temporal scales. The strategic level maintains centralized access to both agents’ states and an exclusive global environmental state $\mathcal{S}^{\text{env}}$ capturing system-wide properties essential for coordination. Strategic guidance, generated every T timesteps, broadcasts to tactical agents who execute distributed policies based on augmented local observations. Tactical rewards accumulate and influence strategic objectives through coupling coefficient λ , creating feedback that grounds strategic planning in tactical realities. An asynchronous protocol with strategic updates every N_f tactical episodes manages non-stationary dynamics, allowing tactical convergence between strategic changes.

Our contributions are threefold. First, we formalize the hierarchical MARL problem with bidirectional coupling and information asymmetry, providing both a general bi-level formulation and tractable additive approximation. Second, we develop the Asynchronous Hierarchical Policy Gradient (AHPG) algorithm managing non-stationarity through temporal separation while maintaining $\mathcal{O}(\log K/\sqrt{K})$ convergence guarantees for the strategic layer. Third, we validate the framework empirically, demonstrating stable bidirectional learning in a cooperative multi-agent foraging task.

CHMAS: 一个耦合的分层多智能体强化学习框架

王东明, 徐杰, 张彦宇, 和任伟

摘要—多智能体强化学习 (MARL) 系统在平衡不同时间尺度上的全局协调与局部执行方面面临基本挑战。本文介绍了耦合的分层多智能体系统 (CHMAS)，这是一个将多智能体决策分解为集中式战略规划和分布式战术执行的新框架，并具有双向信息流。战略层将所有智能体的状态与一个专用的全局环境状态集成，以每隔 T 个时间步生成指导行动，而战术智能体执行由战略指导和本地邻域观察增强的分布式策略。与现有单向控制的分层方法不同，CHMAS 建立了一个反馈机制，其中累积的战术奖励通过耦合系数 λ 影响战略目标，确保战略规划保持与战术可行性一致。为了解决分层学习中固有的非平稳性问题，我们提出了一种异步更新协议，其中战略参数每 N_f 个战术回合更新一次，允许战术策略在战略变化之间收敛到准平稳点。我们提出了一个捕获完整系统动态的通用双层公式和一个便于分析的加性近似。理论分析证明，在标准假设下，该异步方案在 K 次战略更新后，战略层实现了 $(\log K/\sqrt{K})$ 收敛。在多智能体觅食领域的实验验证表明，成功学习了空间分割的探索策略，两层都稳定收敛，尽管存在分层耦合。

I. 介绍

多智能体强化学习 (MARL) 已成为复杂环境中自主协调的基本范式，其应用范围涵盖机器人[1], 交通管理 [2]和资源分配 [3]。尽管取得了显著进展，但在现实世界的部署中仍面临一个关键挑战：在多个时间尺度和空间尺度上协调分布式智能体，同时保持计算的易处理性。

维度的诅咒使得纯粹的集中式方法变得难以处理。对于具有状态空间 $\mathcal{S}_i$ 和动作空间 $\mathcal{A}_i$ 的 N 智能体，联合空间增长为 $\mathcal{O}(\prod_{i=1}^N |\mathcal{S}_i| \cdot |\mathcal{A}_i|)$ [4]。虽然分布式方法通过本地决策 [5], 解决了可扩展性问题，但它们为了长时程目标牺牲了协调质量。像 QMIX [6]和 MADDPG [7] 这样的传统方法利用集中式训练和分布式执行，但假设所有智能体在训练期间都能访问相关的全局信息，这在实践中往往失败

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由于部分可观察性和通信限制 [8]。

此外，多智能体系统本质上跨越多个时间尺度运行。战略决策（编队控制、任务分配）需要包括个体智能体无法获取的环境状态在内的全局信息，这些信息需要在较长时间尺度上运作。战术响应（避碰、局部适应）需要基于即时观察进行快速反应。虽然分层强化学习为多尺度决策提供了一个自然的框架 [9], [10], 但现有的分层多智能体强化学习方法 [11], [12]采用单向控制，其中高层策略决定低层动作，从而错失了战术反馈来指导战略规划的机会。

层级之间的交互产生了复杂的动态学习。战略变化从根本上改变了战术智能体感知的环境，违反了平稳性假设 [8]。相反，演化的战术策略改变了战略规划者所依赖的执行特征。之前的层级多智能体强化学习工作 [13], [14]通过奖励工程或选项学习来解决这个问题，但需要领域专业知识，并且通常无法泛化。

本文介绍了CHMAS，它通过在不同时间尺度上战略层和战术层之间的双向耦合来解决这些限制。战略层保持对智能体状态和独家全局环境状态 $\mathcal{S}^{\text{env}}$ 的集中访问，该状态捕获对协调至关重要的系统级属性。战略

指导，每 T 步生成，向战术智能体广播，这些智能体根据增强的局部观察执行分布式策略。战术奖励通过耦合系数 λ 累积并影响战略目标，形成反馈，使战略规划立足于战术现实。一个异步协议，每 N_f 个战术回合进行战略更新，管理非平稳动态，允许战术在战略变化之间收敛。

我们的贡献有三方面。首先，我们形式化了具有双向耦合和信息不对称的分层MARL问题，提供了通用的双层公式和可处理的加性近似。其次，我们开发了异步分层策略梯度 (AHPG) 算法，通过时间分离管理非平稳性，同时为策略层保持 $\mathcal{O}(\log K/\sqrt{K})$ 收敛保证。第三，我们通过合作多智能体觅食任务的经验验证了该框架，展示了稳定的双向学习。

II. PRELIMINARIES

We consider a cooperative multi-agent system with N agents operating in a shared environment. The system employs a hierarchical architecture with a strategic layer that accesses global information at low temporal frequency and a tactical layer that operates distributedly at high frequency using only local information.

A. Multi-Agent Communication Graph

The tactical agents interact through a time-invariant communication graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ where $\mathcal{V} = \{1, 2, \dots, N\}$ represents the agent set and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ denotes the edge set [15], [16]. For each agent i , the neighborhood $\mathcal{N}_i = \{j \in \mathcal{V} : (i, j) \in \mathcal{E}\}$ is the set of agents that can directly communicate with agent i . This graph constrains information flow at the tactical level, ensuring distributed execution remains scalable.

B. Strategic MDP

Definition 1 (Strategic MDP): The strategic layer evolves as $\mathcal{M}^{\text{str}} = (\mathcal{S}^{\text{str}}, \mathcal{A}^{\text{str}}, P^{\text{str}}, R^{\text{str}}, \gamma_{\text{str}})$, where: (i) the state $\mathcal{S}^{\text{str}} = \mathcal{S}^{\text{env}} \times \prod_{i=1}^N \mathcal{S}_i$ combines the exclusive global environmental state $\mathcal{S}^{\text{env}}$ (resource distributions, global objectives) with all agent states, so that at epoch k the strategic state is $s_k^{\text{str}} = (s_k^{\text{env}}, s_{1,k}, \dots, s_{N,k})$; (ii) actions $\mathcal{A}^{\text{str}} = \prod_{i=1}^N \mathcal{A}_i^{\text{str}}$ provide personalized guidance $a_{i,k}^{\text{str}}$ to each agent, persisting for the interval $[kT, (k+1)T)$; and (iii) the reward integrates global objectives with tactical feedback:

$$R^{\text{str}}(s_k^{\text{str}}, \mathbf{a}_k^{\text{str}}) = R^{\text{global}}(s_k^{\text{env}}, \mathbf{a}_k^{\text{str}}) + \lambda \sum_{t=kT}^{(k+1)T-1} \sum_{i=1}^N r_{i,t}^{\text{tac}}, \quad (1)$$

where $\lambda \in [0, 1]$ controls the influence of tactical execution quality on strategic evaluation.

The strategic policy $\pi^{\text{str}} : \mathcal{S}^{\text{str}} \rightarrow \mathcal{P}(\mathcal{A}^{\text{str}})$ maps complete state observations to guidance distributions, encoding high-level coordination logic that individual agents cannot compute from local information alone.

C. Tactical Multi-Agent MDP

Definition 2 (Tactical MAMDP): Given fixed strategic guidance $\mathbf{a}_k^{\text{str}}$, the tactical layer evolves as $\mathcal{M}^{\text{tac}} = (\{\mathcal{S}_i\}, \{\mathcal{O}_i^{\text{tac}}\}, \{\mathcal{A}_i^{\text{tac}}\}, P^{\text{tac}}, \{R_i^{\text{tac}}\}, \gamma_{\text{tac}})$, where: (i) each agent i observes

$$o_{i,t}^{\text{tac}} = \phi_i(s_{i,t}, \{s_{j,t}\}_{j \in \mathcal{N}_i}, a_{i,k}^{\text{str}}) \in \mathcal{O}_i^{\text{tac}}, \quad (2)$$

combining local state, neighbor states, and strategic guidance ($k = \lfloor t/T \rfloor$); (ii) the joint transition factorizes over the communication graph: $P^{\text{tac}}(\mathbf{s}_{t+1} | \mathbf{s}_t, \mathbf{a}_t^{\text{tac}}) = \prod_{i=1}^N P_i(s_{i,t+1} | s_{i,t}, \{s_{j,t}, a_{j,t}^{\text{tac}}\}_{j \in \mathcal{N}_i \cup \{i\}})$, maintaining tractability as N grows [4]; and (iii) the per-agent reward

$$R_i^{\text{tac}}(s_{i,t}, a_{i,t}^{\text{tac}}, \{s_{j,t}, a_{j,t}^{\text{tac}}\}_{j \in \mathcal{N}_i}, a_{i,k}^{\text{str}}) \quad (3)$$

enables local optimization and neighbor coordination while incorporating strategic objectives.

D. Bidirectional Coupling

Unlike traditional top-down hierarchies [11], CHMAS establishes feedback in both directions. *Downward coupling* (strategic $\rightarrow$ tactical): guidance $a_{i,k}^{\text{str}}$ shapes tactical behavior through observation augmentation (2), reward modulation (3), and temporal persistence over T timesteps. *Upward coupling* (tactical $\rightarrow$ strategic): accumulated rewards $R_k^{\text{tac,cum}} = \sum_{t=kT}^{(k+1)T-1} \sum_{i=1}^N r_{i,t}^{\text{tac}}$ influence strategic rewards via λ in (1), so that strategic decisions account for tactical feasibility. Setting $\lambda = 0$ recovers a purely top-down hierarchy. The framework maintains asymmetric information access: the strategic layer observes complete state s_k^{str} while tactical agents access only $\mathcal{N}_i$, with strategic guidance serving as an information bottleneck [17].

III. PROBLEM FORMULATION

A. General Coupled Formulation

The bidirectional coupling creates a bi-level optimization structure [18], [19]. The strategic objective:

$$J^{\text{str}}(\theta^{\text{str}} | \{\theta_i^{\text{tac}}\}) = \mathbb{E} \left[\sum_{k=0}^{\infty} \gamma_{\text{str}}^k R^{\text{str}}(s_k^{\text{str}}, \mathbf{a}_k^{\text{str}}) \right] \quad (4)$$

depends on tactical policies through (1). Each tactical objective:

$$J_i^{\text{tac}}(\theta_i^{\text{tac}} | \theta^{\text{str}}) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma_{\text{tac}}^t r_{i,t}^{\text{tac}} \right] \quad (5)$$

depends on strategic parameters through (2) and (3).

Definition 3 (Coupled Hierarchical Problem): The coupled optimization problem is the bi-level program:

$$\begin{aligned} \max_{\theta^{\text{str}}} \quad & J^{\text{str}}(\theta^{\text{str}} | \{\theta_i^{\text{tac}*}(\theta^{\text{str}})\}) \\ \text{s.t.} \quad & \theta_i^{\text{tac}*}(\theta^{\text{str}}) = \arg \max_{\theta_i^{\text{tac}}} J_i^{\text{tac}}(\theta_i^{\text{tac}} | \theta^{\text{str}}), \quad \forall i \in \mathcal{V}, \end{aligned}$$

where the strategic optimization accounts for tactical best responses.

Remark 1: We operate in the cooperative MARL setting where agents share a common team objective. Accordingly, the per-agent best responses $\theta_i^{\text{tac}*}$ are aligned with the joint team optimum, and no equilibrium-seeking across agents is required at the tactical level [4].

This Stackelberg structure is computationally intractable in general [20], motivating the approximation below.

B. Additive Approximation for Tractable Optimization

Definition 4 (Additive Hierarchical Problem): Under the additive approximation, the global objective is:

$$J^{\text{global}}(\theta^{\text{str}}, \{\theta_i^{\text{tac}}\}) = J^{\text{str}}(\theta^{\text{str}}) + \sum_{i=1}^N J_i^{\text{tac}}(\theta_i^{\text{tac}}), \quad (6)$$

where each term is evaluated with fixed parameters of the other layer.

Remark 2: Equation (6) is an approximation of the true coupled objective: the cross-layer dependencies in (4)–(5) are not eliminated but handled implicitly through the asynchronous update schedule. The approximation error is

II. 初步知识

我们考虑一个具有 N 个智能体的合作多智能体系统，这些智能体在一个共享环境中运行。该系统采用分层架构，其中战略层以低时间频率访问全局信息，而战术层则以高频率分布式运行，仅使用局部信息。

A. 多智能体通信图

战术智能体通过一个时间不变的通信图 $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ 相互作用，其中 $\mathcal{V} = \{1, 2, \dots, N\}$ 表示智能体集合， $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ 表示边集合 [15], [16]。对于每个智能体 i ，邻域 $\mathcal{N}_i = \{j \in \mathcal{V} : (i, j) \in \mathcal{E}\}$ 是可以直接与智能体 i 通信的智能体集合。这个图限制了战术层面的信息流，确保分布式执行保持可扩展性。

B. 战略MDP

定义1 (战略MDP) : 战略层随 $\mathcal{M}^{\text{str}} = (\mathcal{S}^{\text{str}}, \mathcal{A}^{\text{str}}, P^{\text{str}}, R^{\text{str}}, \gamma_{\text{str}})$ 变化，其中：(i) 状态 $\mathcal{S}^{\text{str}} = \mathcal{S}^{\text{env}} \times \prod_{i=1}^N \mathcal{S}_i$ 结合了排他的全局环境状态 $\mathcal{S}^{\text{env}}$ （资源分布、全局目标）与所有智能体状态，因此在时期 k 战略状态是 $s_k^{\text{str}} = (s_k^{\text{env}}, s_{1,k}, \dots, s_{N,k})$ ；(ii) 动作 $\mathcal{A}^{\text{str}} = \prod_{i=1}^N \mathcal{A}_i^{\text{str}}$ 为每个智能体提供个性化指导 $a_{i,k}^{\text{str}}$ ，持续时间为间隔 $[kT, (k+1)T)$ ；以及 (iii) 奖励整合了全局目标与战术反馈：

$$R^{\text{str}}(s_k^{\text{str}}, \mathbf{a}_k^{\text{str}}) = R^{\text{global}}(s_k^{\text{env}}, \mathbf{a}_k^{\text{str}}) + \lambda \sum_{t=kT}^{(k+1)T-1} \sum_{i=1}^N r_{i,t}^{\text{tac}}, \quad (1)$$

$\lambda \in [0, 1]$ 控制着战术执行质量对战略评估的影响。

战略策略 $\pi^{\text{str}} : \mathcal{S}^{\text{str}} \rightarrow \mathcal{P}(\mathcal{A}^{\text{str}})$ 将完整的状态观察映射到指导分布，编码了个体智能体无法仅从局部信息中计算出的高级协调逻辑。

C. 战术多智能体 MDP

定义2(战术MAMDP): 给定固定的战略指导 $\mathbf{a}_k^{\text{str}}$ ，战术层演变为 $\mathcal{M}^{\text{tac}} = (\{\mathcal{S}_i\}, \{\mathcal{O}_i^{\text{tac}}\}, \{\mathcal{A}_i^{\text{tac}}\}, P^{\text{tac}}, \{R_i^{\text{tac}}\}, \gamma_{\text{tac}})$ ，其中：(i) 每个智能体 i 观察

$$o_{i,t}^{\text{tac}} = \phi_i(s_{i,t}, \{s_{j,t}\}_{j \in \mathcal{N}_i}, a_{i,k}^{\text{str}}) \in \mathcal{O}_i^{\text{tac}}, \quad (2)$$

结合局部状态、邻居状态和战略指导 ($k = \lfloor t/T \rfloor$)；(ii) 联合转移在通信图上分解： $P^{\text{tac}}(\mathbf{s}_{t+1} | \mathbf{s}_t) = \prod_{i=1}^N P_i(s_{i,t+1} | s_{i,t}, \{s_{j,t}, a_{j,t}^{\text{tac}}\}_{j \in \mathcal{N}_i \cup \{i\}}), \mathbf{a}_t^{\text{tac}})$ (...),

随着 N 增长保持可扩展性；以及(iii) 每个智能体的奖励

$$R_i^{\text{tac}}(s_{i,t}, a_{i,t}^{\text{tac}}, \{s_{j,t}, a_{j,t}^{\text{tac}}\}_{j \in \mathcal{N}_i}, a_{i,k}^{\text{str}}) \quad (3)$$

支持局部优化和邻近智能体协调，同时融入战略目标。

D. 双向耦合

与传统自上而下的层级结构不同 [11]，CHMAS建立双向反馈。向下耦合(战略 $\rightarrow$ 战术)：指导 $a_{i,k}^{\text{str}}$ 通过观察增强 (2)、奖励调制 (3) 以及在 T 时间步长上的持续影响，塑造战术行为。 向上耦合 (战术

$$R_k^{\text{tac,cum}} = \sum_{t=kT}^{(k+1)T-1} \sum_{i=1}^N r_{i,t}^{\text{tac}}$$

略)：累积的奖励..通过在 (1) 中影响战略奖励，使得战略决策考虑战术可行性。设置 $\lambda = 0$ 恢复纯粹的层级结构。该框架维持非对称信息访问：战略层观察完整状态 s_k^{str} ，而战术智能体仅访问 $\mathcal{N}_i$ ，战略指导作为信息瓶颈 [17]。

III. 问题公式化

A. 一般耦合公式

双向耦合创建了一个双层优化结构 [18], [19]。战略目标：

$$J^{\text{str}}(\theta^{\text{str}} | \{\theta_i^{\text{tac}}\}) = \mathbb{E} \left[\sum_{k=0}^{\infty} \gamma_{\text{str}}^k R^{\text{str}}(s_k^{\text{str}}, \mathbf{a}_k^{\text{str}}) \right] \quad (4)$$

依赖于通过 (1) 的战术策略。每个战术目标：

$$J_i^{\text{tac}}(\theta_i^{\text{tac}} | \theta^{\text{str}}) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma_{\text{tac}}^t r_{i,t}^{\text{tac}} \right] \quad (5)$$

依赖于通过 (2) 和 (3) 的战略参数。

定义3(耦合分层问题): 耦合优化问题是双层规划：

$$\begin{aligned} \max_{\theta^{\text{str}}} \quad & J^{\text{str}}(\theta^{\text{str}} | \{\theta_i^{\text{tac}*}(\theta^{\text{str}})\}) \\ \text{s.t.} \quad & \theta_i^{\text{tac}*}(\theta^{\text{str}}) = \arg \max_{\theta_i^{\text{tac}}} J_i^{\text{tac}}(\theta_i^{\text{tac}} | \theta^{\text{str}}), \quad \forall i \in \mathcal{V}, \end{aligned}$$

其中战略优化考虑了战术最佳反应。

注释1: 我们在合作多智能体强化学习(MARL)环境中运行，智能体共享一个共同团队目标。因此，每个智能体的最佳反应 $\theta_i^{\text{tac}*}$ 与联合团队最优值一致，并且在战术层面 [4]不需要跨智能体的均衡寻求。

这种Stackelberg结构在一般情况下是计算上不可行的 [20],，从而引出了下面的近似方法。

B. 可行优化加性近似

定义4(加性分层问题): 在加性近似下，全局目标是：

$$J^{\text{global}}(\theta^{\text{str}}, \{\theta_i^{\text{tac}}\}) = J^{\text{str}}(\theta^{\text{str}}) + \sum_{i=1}^N J_i^{\text{tac}}(\theta_i^{\text{tac}}), \quad (6)$$

其中每个项都使用其他层的固定参数进行评估。

备注2: 公式(6)是真实耦合目标的近似：在(4)–(5)中的跨层依赖并非被消除，而是通过异步更新调度隐式处理。近似误差由

controlled by the tactical optimality gap δ_k which will be defined in Assumption 2 at each epoch, which vanishes as learning progresses.

The separable gradient structure,

$$\nabla_{\theta} J^{\text{global}} = [\nabla_{\theta^{\text{str}}} J^{\text{str}}, \nabla_{\theta^{\text{tac}}} J^{\text{tac}}]^T, \quad (7)$$

allows updates to be treated as coordinate steps in the joint parameter space, enabling rigorous convergence analysis via two-timescale stochastic approximation [21], [22].

IV. ASYNCHRONOUS LEARNING ALGORITHM

A. Algorithm Design Principles

AHPG leverages three principles for stable hierarchical learning.

Temporal separation. Strategic decisions occur every T timesteps and persist throughout the interval. Strategic parameters follow a decaying step size schedule $\eta_k^{\text{str}} = \alpha/\sqrt{k}$ and update only every N_f tactical episodes, operating at a slower effective timescale than tactical learning. This allows tactical policies to converge to near-optimal responses between strategic changes, transforming the non-stationary hierarchical problem into alternating stationary subproblems.

Structured information flow. Downward flow provides coordination signals through strategic guidance without requiring centralized control at each timestep. Upward flow aggregates tactical performance $R_k^{\text{tac}} = \sum_{t,i} r_{i,t}^{\text{tac}}$ into strategic rewards, keeping high-level plans grounded in execution realities. The strategic layer maintains exclusive access to s_k^{env} , preserving the information asymmetry that enables scalable distributed execution.

Distributed tactical learning. Each tactical agent maintains its own policy π_i^{tac} and experience buffer $\mathcal{D}_i^{\text{tac}}$, updating parameters using only local gradients without inter-agent communication during learning, so that tactical computation scales linearly with N .

B. Algorithm Specification

Algorithm 1 presents the complete AHPG procedure. Strategic guidance is generated at the start of each new strategic epoch (every N_f episodes), and strategic parameters are updated at the end of each epoch using the accumulated tactical performance. Tactical parameters update every episode.

During the N_f episodes between strategic updates, tactical agents operate in a stationary environment with fixed guidance, converging toward near-optimal policies. This ensures strategic updates observe stable tactical performance that accurately estimates the tactical contribution to strategic rewards.

For learning components, the strategic layer employs centralized policy gradient algorithms (e.g., PPO [23]) leveraging complete state access, while tactical agents use distributed policy gradient or actor-critic variants operating on local observations. The framework naturally extends to continuous control through continuous guidance vectors and to partially observable settings through recurrent policies.

Algorithm 1 Asynchronous Hierarchical Policy Gradient (AHPG)

Require: Initial parameters $\theta_1^{\text{str}}, \{\theta_{i,0}^{\text{tac}}\}$

Require: Hyperparameters $T, N_f, \alpha, \eta^{\text{tac}}, \lambda, c$

```

1: Initialize episode counter  $e \leftarrow 0$ , strategic epoch  $k \leftarrow 1$ 

2: for episode  $e = 1, 2, \dots$  do
3:   // Strategic Planning Phase
4:   if  $e \bmod N_f = 1$  then
5:     Observe full state  $s_k^{\text{str}} = (s_k^{\text{env}}, \mathbf{s}_k)$ 
6:     Generate  $\mathbf{a}_k^{\text{str}} \sim \pi^{\text{str}}(s_k^{\text{str}}; \theta_k^{\text{str}})$ 
7:     Broadcast  $a_{i,k}^{\text{str}}$  to each agent  $i$ 
8:   end if
9:   // Tactical Execution Phase
10:  for  $t = 0$  to  $T - 1$  do
11:    for agent  $i \in \mathcal{V}$  in parallel do
12:       $o_{i,t}^{\text{tac}} = \phi_i(s_{i,t}, \{s_{j,t}\}_{j \in \mathcal{N}_i}, a_{i,k}^{\text{str}})$ 
13:       $a_{i,t}^{\text{tac}} \sim \pi_i^{\text{tac}}(o_{i,t}^{\text{tac}}, \theta_{i,e}^{\text{tac}})$ 
14:      Store  $(o_{i,t}^{\text{tac}}, a_{i,t}^{\text{tac}}, r_{i,t}^{\text{tac}})$  in  $\mathcal{D}_i^{\text{tac}}$ 
15:    end for
16:  end for
17:  // Tactical Policy Update
18:  for agent  $i \in \mathcal{V}$  in parallel do
19:     $g_i^{\text{tac}} = \nabla_{\theta^{\text{tac}}} J_i^{\text{tac}}$  using  $\mathcal{D}_i^{\text{tac}}$ 
20:     $\theta_{i,e+1}^{\text{tac}} = \theta_{i,e}^{\text{tac}} + \eta^{\text{tac}} g_i^{\text{tac}}$ 
21:  end for
22:  // Strategic Update
23:  if  $e \bmod N_f = 0$  then
24:     $R_k^{\text{tac}} = \sum_{t=kT}^{(k+1)T-1} \sum_{i=1}^N r_{i,t}^{\text{tac}}$ 
25:     $r_k^{\text{str}} = R^{\text{global}}(s_k^{\text{env}}, \mathbf{a}_k^{\text{str}}) + \lambda R_k^{\text{tac}}$ 
26:     $g^{\text{str}} = \nabla_{\theta^{\text{str}}} J^{\text{str}}$  using  $\mathcal{D}^{\text{str}}$ 
27:     $\theta_{k+1}^{\text{str}} = \theta_k^{\text{str}} + (\alpha/\sqrt{k}) g^{\text{str}}$ 
28:     $k \leftarrow k + 1$ 
29:  end if
30: end for
```

V. CONVERGENCE ANALYSIS

We establish convergence for AHPG under the additive approximation $J^{\text{global}}(\theta) = J^{\text{str}}(\theta^{\text{str}}) + \sum_{i=1}^N J_i^{\text{tac}}(\theta_i^{\text{tac}})$.

Assumption 1: 1) **Smoothness:** J^{str} and each J_i^{tac} are L -smooth with L -Lipschitz gradients.

- 2) **Boundedness:** $J^{\text{str}} \leq J^{\text{str}*}$ and $J_i^{\text{tac}} \leq J_i^{\text{tac}*}$ for all i .
- 3) **Bounded variance:** Stochastic gradient estimates satisfy $\mathbb{E}[\|g_k^{\text{str}} - \nabla J_k^{\text{str}}\|^2] \leq \sigma_{\text{str}}^2$ and $\mathbb{E}[\|g_{i,e}^{\text{tac}} - \nabla J_i^{\text{tac}}\|^2] \leq \sigma_{\text{tac}}^2$.
- 4) **Biased strategic gradient:** $\mathbb{E}[g_k^{\text{str}} \mid \theta_k^{\text{str}}] = \nabla J^{\text{str}}(\theta_k^{\text{str}}) + b_k$, where b_k captures the error from evaluating strategic gradients with non-converged tactical parameters.

Assumption 2: 1) **PL condition:** Each J_i^{tac} satisfies the μ -PL inequality: $\|\nabla J_i^{\text{tac}}(\theta)\|^2 \geq 2\mu(J_i^{\text{tac}*} - J_i^{\text{tac}}(\theta))$.

- 2) **Warm-start:** At the beginning of each strategic epoch k , tactical agents are initialized from the converged state of the previous epoch. The initial optimality gap satisfies $\mathbb{E}[J_i^{\text{tac}*} - J_i^{\text{tac}}(\theta_{i,kN_f}^{\text{tac}})] \leq \delta_k$, where $\delta_k = \mathcal{O}(\eta_k^{\text{str}}) = \mathcal{O}(1/\sqrt{k})$, reflecting the magnitude of the

战术最优性差距 δ_k 控制，该差距将在假设2中在每个 epoch定义，随着学习进展而消失。

可分离的梯度结构，

$$\nabla_{\theta} J^{\text{global}} = [\nabla_{\theta^{\text{str}}} J^{\text{str}}, \nabla_{\theta^{\text{tac}}} J^{\text{tac}}]^T, \quad (7)$$

允许将更新视为联合参数空间中的坐标步长，从而通过双时间尺度随机近似 [21], [22] 实现严格的收敛分析。

IV. 异步学习算法

A. 算法设计原则

AHPG 利用三种原则进行稳定的分层学习。

时间分离。 战略决策每 T 步发生并持续整个间隔。战略参数遵循衰减步长调度 $\eta_k^{\text{str}} = \alpha/\sqrt{k}$ ，并且仅每 N_f 个战术回合更新，以比战术学习更慢的有效时间尺度运行。这允许战术策略在战略变化之间收敛到近似最优响应，将非平稳的分层问题转化为交替的平稳子问题。

结构化信息流。 向下流通过战略指导提供协调信号，而无需在每个时间步长进行集中控制。向上流将战术性能 $R_k^{\text{tac}} = \sum_{t,i} r_{i,t}^{\text{tac}}$ 、 t 聚合为战略奖励，使高层计划根植于执行现实。战略层保持对 s_k^{env} 的独占访问权，保留信息不对称，从而实现可扩展的分布式执行。

分布式战术学习。 每个战术智能体维护自己的策略 π_i^{tac} 和经验缓冲区 $\mathcal{D}_i^{\text{tac}}$ ，在学习过程中仅使用局部梯度更新参数，无需智能体间通信，使得战术计算与 N 线性扩展。

B. 算法规范

算法1展示了完整的AHPG流程。在每个新的策略周期开始时（每 N_f 个回合），生成战略指导，并在每个周期的结束时使用累积的战术性能更新战略参数。战术参数每回合更新一次。

在战略更新之间的 N_f 个回合中，战术智能体在一个具有固定指导的静止环境中运行，并趋近于近最优策略。这确保了战略更新观察到稳定的战术性能，该性能准确估计了战术对战略奖励的贡献。

对于学习组件，策略层采用集中式策略梯度算法（例如PPO [23]），利用完整的全局状态信息，而战术智能体则使用分布式策略梯度或Actor-Critic变体，在局部观察上进行操作。该框架通过连续指导向量自然扩展到连续控制，通过循环策略扩展到部分可观察环境。

算法 (AHPG)	1 异步分层策略	梯度
要求:	初始参数 $\theta_1^{\text{str}}, \{\theta_{i,0}^{\text{tac}}, 0\}$	
需要:	超参数 $T, N_f, \alpha, \eta^{\text{tac}}, \lambda, c$	
	1: 初始化回合计数器 $e \leftarrow 0$, 策略性轮次 $k \leftarrow 1$	
	2: 对于 回合 $e = 1, 2, \dots$ 执行	
	3: // 策略规划阶段	
	4: if $e \bmod N_f = 1$ then	
	5: 观察完整状态 $s_k^{\text{str}} = (s_k^{\text{env}}, \mathbf{s}_k)$	
	6: 生成一个 $\mathbf{a}_k^{\text{str}} \sim \pi^{\text{str}}(s_k^{\text{str}}; \theta_k^{\text{str}})$	
	7: 向每个智能体 i 广播 $a_{i,k}^{\text{str}}$	
	8: end if	
	9: // 战术执行阶段	
	10: for $t = 0$ 到 $T - 1$ do	
	11: 对于 智能体 $i \in \mathcal{V}$ 并行 执行	
	12: $o_{i,t}^{\text{tac}} = \phi_i(s_{i,t}, \{s_{j,t}\}_{j \in \mathcal{N}_i}, a_{i,k}^{\text{str}})$	
	13: $a_{i,t}^{\text{tac}} \sim \pi_i^{\text{tac}}(o_{i,t}^{\text{tac}}, \theta_{i,e}^{\text{tac}})$	
	14: 存储 $(o_{i,t}^{\text{tac}}, a_{i,t}^{\text{tac}}, r_{i,t}^{\text{tac}})$ 在 $\mathcal{D}_i^{\text{tac}}$	
	15: 结束 for	
	16: 结束 for	
	17: // 战术策略更新	
	18: for 智能体 $i \in \mathcal{V}$ 并行 执行	
	19: $g_i^{\text{tac}} = \nabla_{\theta^{\text{tac}}} J_i^{\text{tac}}$ 使用 $\mathcal{D}_i^{\text{tac}}$	
	20: $\theta_{i,e+1}^{\text{tac}} = \theta_{i,e}^{\text{tac}} + \eta^{\text{tac}} g_i^{\text{tac}}$	
	21: 结束for	
	22: // 战略更新	
	23: 如果 $e \bmod N_f = 0$ 则	
	24: $R_k^{\text{tac}} = \sum_{t=kT}^{(k+1)T-1} \sum_{i=1}^N r_{i,t}^{\text{tac}}$	
	25: $r_k^{\text{str}} = R^{\text{global}}(s_k^{\text{env}}, \mathbf{a}_k^{\text{str}}) + \lambda R_k^{\text{tac}}$	
	26: $g^{\text{str}} = \nabla_{\theta^{\text{str}}} J^{\text{str}}$ 使用 $\mathcal{D}^{\text{str}}$	
	27: $\theta_{k+1}^{\text{str}} = \theta_k^{\text{str}} + (\alpha/\sqrt{k}) g^{\text{str}}$	
	28: $k \leftarrow k + 1$	
	29: 结束 if	
	30: 结束 for	

V. 收敛分析

$J^{\text{global}}(\theta) = J^{\text{str}}(\theta^{\text{str}}) + \sum_{i=1}^N J_i^{\text{tac}}(\theta_i^{\text{tac}})$ 下建立了 AHPG 的收敛性。假设 1: 1) **光滑性:** J^{str} 和每个 J_i^{tac} 是 L -光滑的，梯度具有 L -Lipschitz 性质。2) **有界性:** $J^{\text{str}} \leq J^{\text{str}*}$ 和 $J_i^{\text{tac}} \leq J_i^{\text{tac}*}$ 对所有 i 有界。3) **方差有界:** 随机梯度估计满足

$$\mathbb{E}[\|g_k^{\text{str}} - \nabla J_k^{\text{str}}\|^2] \leq \sigma_{\text{str}}^2 \quad \mathbb{E}[\|g_{i,e}^{\text{tac}} - \nabla J_i^{\text{tac}}\|^2] \leq \sigma_{\text{tac}}^2 \quad \text{和} \quad 。$$

4) **偏差策略梯度:** $\mathbb{E}[g_k^{\text{str}} \mid \theta_k^{\text{str}}] = \nabla J^{\text{str}}(\theta_k^{\text{str}}) + b_k$ ，其中 b_k 捕捉了使用未收敛的战术参数评估策略梯度时的误差。假设 2: 1) **PL 条件:** 每个 J_i^{tac} 满足 μ -PL 不等式: $\|\nabla J_i^{\text{tac}}(\theta)\|^2 \geq 2\mu(J_i^{\text{tac}*} - J_i^{\text{tac}}(\theta))$ 。2) **热启动:** 在每个策略周期的开始，战术智能体从上一周期的收敛状态初始化。初始最优性差距满足

$$\mathbb{E}[J_i^{\text{tac}*} - J_i^{\text{tac}}(\theta_{i,kN_f}^{\text{tac}})] \leq \delta_k \quad \delta_k = (.), \quad \text{其}$$

$$\mathcal{O}(\eta_k^{\text{str}}) = \mathcal{O}(1/\sqrt{k})$$

中 $(.)$ ，反映了连续周期期间策略参数变化的幅度。

strategic parameter change between successive epochs.

- 3) **Round-robin updates:** Within each tactical phase, agents update sequentially with others’ parameters held fixed, so that the PL condition applies to each per-agent landscape independently.

Remark 3: The PL condition in Assumption 2 holds for log-linear (softmax) policies in tabular settings [24] and locally for over-parameterized neural networks near stationary points. The round-robin assumption is a standard device to decouple per-agent landscapes; it can be relaxed to simultaneous updates at the cost of a more involved coupling analysis.

Assumption 3 (Coupling structure): The strategic gradient is L_b -Lipschitz in the tactical parameters:

$$\|\nabla J^{\text{str}}(\theta^{\text{str}}, \theta^{\text{tac}}) - \nabla J^{\text{str}}(\theta^{\text{str}}, \theta^{\text{tac}*})\| \leq L_b \|\theta^{\text{tac}} - \theta^{\text{tac}*}\|.$$

Theorem 1 (AHPG Convergence): Under Assumptions 1–3, with $\eta_k^{\text{str}} = \alpha/\sqrt{k}$ where $\alpha \leq 1/(8L)$, $\eta^{\text{tac}} = \beta/\sqrt{K}$ where $\beta < 1/(2L)$, and $N_f = c\sqrt{K}$ for a constant $c > 0$, after K strategic updates AHPG achieves:

$$\min_{k \in \{1, \dots, K\}} \mathbb{E}[\|\nabla J^{\text{str}}(\theta_k^{\text{str}})\|^2] = \mathcal{O}\left(\frac{\log K}{\sqrt{K}}\right),$$

$$\frac{1}{KN_f} \sum_{k=1}^K \sum_{e=kN_f}^{(k+1)N_f-1} \sum_{i=1}^N \mathbb{E}[\|\nabla J_i^{\text{tac}}(\theta_{i,e}^{\text{tac}})\|^2] = \mathcal{O}\left(\frac{1}{\sqrt{K}}\right).$$

Proof: **Strategic layer.** By L -smoothness of J^{str} and the update $\theta_{k+1}^{\text{str}} = \theta_k^{\text{str}} + \eta_k^{\text{str}} g_k^{\text{str}}$:

$$J_{k+1}^{\text{str}} \geq J_k^{\text{str}} + \eta_k^{\text{str}} \langle \nabla J_k^{\text{str}}, g_k^{\text{str}} \rangle - \frac{L}{2} (\eta_k^{\text{str}})^2 \|g_k^{\text{str}}\|^2.$$

Taking expectation with $\mathbb{E}[g_k^{\text{str}}] = \nabla J_k^{\text{str}} + b_k$, we bound the two terms. For the inner product, applying Young’s inequality $\langle a, b \rangle \geq -\frac{1}{4}\|a\|^2 - \|b\|^2$:

$$\mathbb{E}[\langle \nabla J_k^{\text{str}}, g_k^{\text{str}} \rangle] \geq \frac{3}{4} \|\nabla J_k^{\text{str}}\|^2 - \|b_k\|^2.$$

For the squared norm, using $\mathbb{E}[\|g_k^{\text{str}}\|^2] \leq \|\nabla J_k^{\text{str}} + b_k\|^2 + \sigma_{\text{str}}^2 \leq 2\|\nabla J_k^{\text{str}}\|^2 + 2\|b_k\|^2 + \sigma_{\text{str}}^2$:

$$\mathbb{E}[J_{k+1}^{\text{str}}] \geq J_k^{\text{str}} + \left(\frac{3}{4}\eta_k^{\text{str}} - L(\eta_k^{\text{str}})^2\right) \|\nabla J_k^{\text{str}}\|^2 - \left(\eta_k^{\text{str}} + L(\eta_k^{\text{str}})^2\right) \|b_k\|^2 - \frac{L}{2} (\eta_k^{\text{str}})^2 \sigma_{\text{str}}^2.$$

Since $\eta_k^{\text{str}} \leq 1/(8L)$, we have $L(\eta_k^{\text{str}})^2 \leq \eta_k^{\text{str}}/8$, so $\frac{3}{4}\eta_k^{\text{str}} - L(\eta_k^{\text{str}})^2 \geq \frac{5}{8}\eta_k^{\text{str}} \geq \frac{1}{8}\eta_k^{\text{str}}$ and $\eta_k^{\text{str}} + L(\eta_k^{\text{str}})^2 \leq \frac{9}{8}\eta_k^{\text{str}} \leq 2\eta_k^{\text{str}}$. Rearranging and telescoping over $k = 1, \dots, K$:

$$\frac{1}{8} \sum_{k=1}^K \eta_k^{\text{str}} \mathbb{E}[\|\nabla J_k^{\text{str}}\|^2] \leq \Delta J^{\text{str}} + 2 \sum_{k=1}^K \eta_k^{\text{str}} \|b_k\|^2 + \frac{L\sigma_{\text{str}}^2}{2} \sum_{k=1}^K (\eta_k^{\text{str}})^2, \quad (8)$$

where $\Delta J^{\text{str}} = J^{\text{str}*} - J_1^{\text{str}}$.

Bias bound. By Assumption 3, $\|b_k\|^2 \leq L_b^2 \|\theta_{(k+1)N_f}^{\text{tac}} - \theta^{\text{tac}*}\|^2$. Running $N_f = c\sqrt{K}$ SGD steps under the PL condition (Assumption 2) with step size $\beta/\sqrt{K}$ and warm-start gap δ_k yields contraction factor $\rho = (1 - 2\mu\beta/\sqrt{K})^{c\sqrt{K}} \rightarrow e^{-2\mu\beta c} < 1$, so the residual tactical deviation satisfies $\|b_k\|^2 = \mathcal{O}(\delta_k + 1/\sqrt{K}) = \mathcal{O}(1/\sqrt{k})$.

Completing the strategic bound. With $\eta_k^{\text{str}} = \alpha/\sqrt{k}$:

$$\sum_{k=1}^K \eta_k^{\text{str}} \|b_k\|^2 = \mathcal{O}\left(\alpha \sum_{k=1}^K \frac{1}{\sqrt{k}} \cdot \frac{1}{\sqrt{k}}\right) = \mathcal{O}(\log K),$$

$$\sum_{k=1}^K (\eta_k^{\text{str}})^2 = \alpha^2 \sum_{k=1}^K \frac{1}{k} = \mathcal{O}(\log K).$$

Dividing (8) by $\sum_{k=1}^K \eta_k^{\text{str}} = \mathcal{O}(\alpha\sqrt{K})$ gives $\min_k \mathbb{E}[\|\nabla J_k^{\text{str}}\|^2] = \mathcal{O}(\log K/\sqrt{K})$.

Tactical layer. Within epoch k , θ_k^{str} is fixed, so J_i^{tac} is stationary. Standard SGD under the μ -PL condition over N_f steps gives per-epoch average gradient norm $\mathcal{O}(\delta_k + 1/\sqrt{K})$. Averaging over K epochs and using $K^{-1} \sum_{k=1}^K \delta_k = K^{-1} \sum_{k=1}^K \mathcal{O}(k^{-1/2}) = \mathcal{O}(K^{-1/2})$, the overall tactical average is $\mathcal{O}(1/\sqrt{K})$, completing the proof. ■

Remark 4: Theorem 1 is established for policy gradient implementations. The DQN-based experiments in Section VI serve as an empirical proof of concept; extending the convergence analysis to Q-learning variants is left for future work.

VI. EXPERIMENTAL VALIDATION

A. Experiment Setting

We evaluate CHMAS on a 25×25 GridWorld environment with $N = 4$ agents performing cooperative foraging, instantiating the key features of hierarchical multi-agent coordination: information asymmetry, temporal abstraction, and bidirectional coupling.

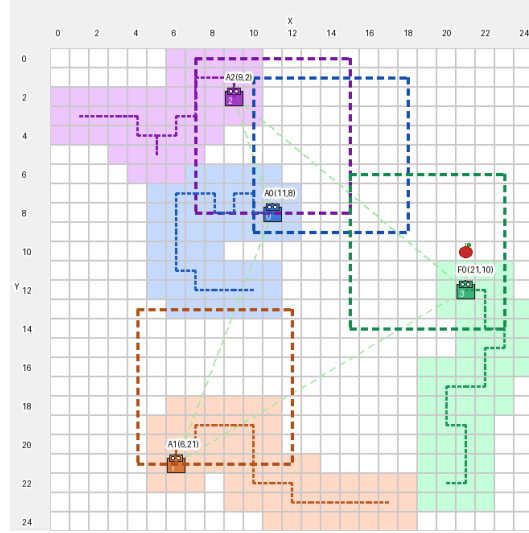


Fig. 1: The 25×25 GridWorld environment. Four agents (A0-blue, A1-orange, A2-purple, A3-green) operate within their assigned 9×9 regions (dashed boxes) determined by strategic guidance. Colored areas indicate visited cells tracked in the global state s^{env} , while agents pursue local resources (red apple).

The global environmental state $s^{\text{env}} \in \{0, 1\}^{625}$ tracks cell visitation, representing system-wide coverage that individual agents cannot observe. The strategic layer accesses the complete state $s^{\text{str}} = (s^{\text{env}}, s_1, s_2, s_3, s_4)$, while each tactical

continuous period strategy parameter change amplitude.

- 3) **轮询更新:** 在每个战术阶段内, 智能体按顺序更新, 同时其他智能体的参数保持固定, 因此PL条件可独立应用于每个智能体的景观。

备注3: 假设2中的PL条件适用于表格环境中的对数线性 (softmax) 策略 [24], 以及局部适用于过参数化神经网络在稳定点附近。轮询假设是一种标准的解耦每个智能体景观的机制; 它可以放宽为同时更新, 但代价是更复杂的耦合分析。

假设3 (耦合结构): 策略梯度在战术参数上是 L_b -Lipschitz的:

$$\|\nabla J^{\text{str}}(\theta^{\text{str}}, \theta^{\text{tac}}) - \nabla J^{\text{str}}(\theta^{\text{str}}, \theta^{\text{tac}*})\| \leq L_b \|\theta^{\text{tac}} - \theta^{\text{tac}*}\|.$$

定理1 (AHPG收敛性): 在假设1-3下, 当 $\eta_k^{\text{str}} = \alpha/\sqrt{k}$ 且 $\alpha \leq 1/(8L)$, $\eta^{\text{tac}} = \beta/\sqrt{K}$ 且 $\beta < 1/(2L)$, 以及 $N_f = c\sqrt{K}$ 对常数 $c > 0$ 成立时, 经过 K 次策略更新后, AHPG实现:

$$\min_{k \in \{1, \dots, K\}} \mathbb{E}[\|\nabla J^{\text{str}}(\theta_k^{\text{str}})\|^2] = \mathcal{O}\left(\frac{\log K}{\sqrt{K}}\right),$$

$$\frac{1}{KN_f} \sum_{k=1}^K \sum_{e=kN_f}^{(k+1)N_f-1} \sum_{i=1}^N \mathbb{E}[\|\nabla J_i^{\text{tac}}(\theta_{i,e}^{\text{tac}})\|^2] = \mathcal{O}\left(\frac{1}{\sqrt{K}}\right).$$

证明: **策略层.** 通过 L -平滑性 J^{str} 和更新 $\theta_{k+1}^{\text{str}} = \theta_k^{\text{str}} + \eta_k^{\text{str}} g_k^{\text{str}}$:

$$J_{k+1}^{\text{str}} \geq J_k^{\text{str}} + \eta_k^{\text{str}} \langle \nabla J_k^{\text{str}}, g_k^{\text{str}} \rangle - \frac{L}{2} (\eta_k^{\text{str}})^2 \|g_k^{\text{str}}\|^2.$$

对 $\mathbb{E}[g_k^{\text{str}}] = \nabla J_k^{\text{str}} + b_k$ 取期望, 我们界定了这两个项。对于内积, 应用杨氏不等式 $\langle a, b \rangle \geq -\frac{1}{4}\|a\|^2 - \|b\|^2$:

$$\mathbb{E}[\langle \nabla J_k^{\text{str}}, g_k^{\text{str}} \rangle] \geq \frac{3}{4} \|\nabla J_k^{\text{str}}\|^2 - \|b_k\|^2.$$

对于平方范数, 使用 $\mathbb{E}[\|g_k^{\text{str}}\|^2] \leq \|\nabla J_k^{\text{str}} + b_k\|^2 + \sigma_{\text{str}}^2 \leq 2\|\nabla J_k^{\text{str}}\|^2 + 2\|b_k\|^2 + \sigma_{\text{str}}^2$:

$$\mathbb{E}[J_{k+1}^{\text{str}}] \geq J_k^{\text{str}} + \left(\frac{3}{4}\eta_k^{\text{str}} - L(\eta_k^{\text{str}})^2\right) \|\nabla J_k^{\text{str}}\|^2 - \left(\eta_k^{\text{str}} + L(\eta_k^{\text{str}})^2\right) \|b_k\|^2 - \frac{L}{2} (\eta_k^{\text{str}})^2 \sigma_{\text{str}}^2.$$

由于 $\eta_k^{\text{str}} \leq 1/(8L)$, $L(\eta_k^{\text{str}})^2 \leq \eta_k^{\text{str}}/8$, $\frac{3}{4}\eta_k^{\text{str}} - L(\eta_k^{\text{str}})^2 \geq \frac{5}{8}\eta_k^{\text{str}} \geq \frac{1}{8}\eta_k^{\text{str}}$ 和 $\eta_k^{\text{str}} + L(\eta_k^{\text{str}})^2 \leq \frac{9}{8}\eta_k^{\text{str}} \leq 2\eta_k^{\text{str}}$ 。重新排列并通过 $k = 1, \dots, K$ 进行望远镜求和:

$$\frac{1}{8} \sum_{k=1}^K \eta_k^{\text{str}} \mathbb{E}[\|\nabla J_k^{\text{str}}\|^2] \leq \Delta J^{\text{str}} + 2 \sum_{k=1}^K \eta_k^{\text{str}} \|b_k\|^2 + \frac{L\sigma_{\text{str}}^2}{2} \sum_{k=1}^K (\eta_k^{\text{str}})^2, \quad (8)$$

在 $\Delta J^{\text{str}} = J^{\text{str}*} - J_1^{\text{str}}$ 处。

偏差界限. 根据假设3 $\|b_k\|^2 \leq L_b^2 \|\theta_{(k+1)N_f}^{\text{tac}} - \theta^{\text{tac}*}\|^2$ 。在 PL 条件 (假设2) 下使用步长 $\beta/\sqrt{K}$ 和预热间隔 δ_k 运行 SGD 步骤产生收缩因子 $\rho = (1 - 2\mu\beta/\sqrt{K})^{c\sqrt{K}} \rightarrow e^{-2\mu\beta c} < 1$, 因此残差策略偏差满足 $\|b_k\|^2 = \mathcal{O}(\delta_k + 1/\sqrt{K}) = \mathcal{O}(1/\sqrt{k})$ 。

完成策略界限. 使用 $\eta_k^{\text{str}} = \alpha/\sqrt{k}$:

$$\sum_{k=1}^K \eta_k^{\text{str}} \|b_k\|^2 = \mathcal{O}\left(\alpha \sum_{k=1}^K \frac{1}{\sqrt{k}} \cdot \frac{1}{\sqrt{k}}\right) = \mathcal{O}(\log K),$$

$$\sum_{k=1}^K (\eta_k^{\text{str}})^2 = \alpha^2 \sum_{k=1}^K \frac{1}{k} = \mathcal{O}(\log K).$$

将 (8) 除以 $\sum_{k=1}^K \eta_k^{\text{str}} = \mathcal{O}(\alpha\sqrt{K})$ 得到 $\min_k \mathbb{E}[\|\nabla J_k^{\text{str}}\|^2] = \mathcal{O}(\log K/\sqrt{K})$ 。

战术层. 在每个 epoch k 中, θ_k^{str} 是固定的, 因此 J_i^{tac} 是稳定的。在 μ -PL 条件下, 标准 SGD 经过 N_f 步后, 得到每个 epoch 的平均梯度范数 $\mathcal{O}(\delta_k + 1/\sqrt{K})$ 。对 $K^{-1} \sum_{k=1}^K \delta_k = \text{一个 epoch 进行平均, 并使用 } ()$, 整体战术平均值为 $\mathcal{O}(1/\sqrt{K})$, 从而完成证明。

备注4: 定理1是针对策略梯度实现的。第六节的基于DQN的实验是概念验证的实证证明; 将收敛分析扩展到Q-learning变体是未来工作的内容。

VI. 实验验证

A. 实验设置

我们在一个包含 25×25 智能体的 25×25 网格世界环境中评估 CHMAS, 这些智能体执行合作觅食任务, 实例化了分层多智能体协调的关键特征: 信息不对称、时间抽象和双向耦合。

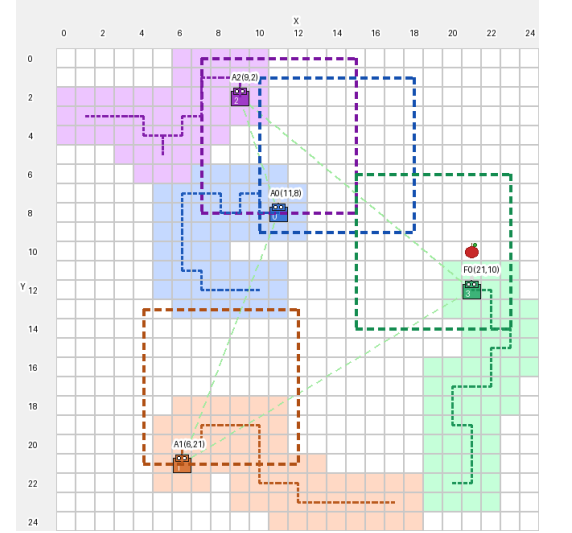


图1: 25×25 网格世界环境。四个智能体 (A0-蓝色, A1-橙色, A2-紫色, A3-绿色) 在其被策略性指导确定的分配 9×9 区域内操作 (虚线框)。彩色区域表示在全局状态 s^{env} 中追踪的已访问单元格, 而智能体则追求本地资源 (红色苹果)。

全局环境状态 $s^{\text{env}} \in \{0, 1\}^{625}$ 追踪单元访问情况, 代表着系统范围内的覆盖范围, 单个智能体无法观察到。策略层访问完整状态 $s^{\text{str}} = (s^{\text{env}}, s_1, s_2, s_3, s_4)$, 而每个战术

agent i observes only local information $o_i^{\text{tac}} \in \mathbb{R}^{20}$ comprising its position, nearby resources, and strategic guidance.

Strategic actions assign each agent a 9×9 region center $a_i^{\text{str}} = (g_i^x, g_i^y) \in [4, 20]^2$ at the start of each epoch ($T = 5$ timesteps). Tactical agents select discrete actions $a_i^{\text{tac}} \in \{0, 1, 2, 3, 4, 5\}$ (movement, collection, wait).

The reward structure implements bidirectional coupling: tactical agents receive $r_{i,t}^{\text{tac}} = r_{i,t}^{\text{collect}} - 0.01 - 0.01 \cdot \mathbf{1}[\text{outside region}]$, while the strategic layer exclusively receives global coverage reward $R_k^{\text{global}} = 0.01 \times |\text{newly visited cells}|$ computed from s^{env} . Strategic rewards integrate both: $R_k^{\text{str}} = R_k^{\text{global}} + \lambda \sum_{t,i} r_{i,t}^{\text{tac}}$ with $\lambda = 0.5$.

The implementation is publicly available at https://github.com/EricDmWang/Hierarchy_RL.git.

B. Results and Analysis

We implement the tactical layer using Multi-Agent DQN (MA-DQN) with parameter sharing, dueling architecture, double Q-learning, and prioritized experience replay. The strategic layer employs a centralized DQN on the global state. Training runs for 400 episodes with 5 independent seeds.

Figure 2 shows the tactical layer’s learning progression. Q-values (a) grow rapidly before stabilizing near episode 300. Returns (b) improve from approximately -80 to -10 , showing agents learn efficient resource collection while respecting region constraints. Critic loss (c) and policy loss (d) peak early during exploration and then decrease as policies stabilize.

Figure 3 shows the strategic layer’s asynchronous learning with updates every $N_f = 5$ episodes. Strategic Q-values (a) improve from -8 to near 0, and strategic returns (b) increase from -10 to approximately 15, demonstrating successful optimization of global coverage objectives. The delayed onset of learning (first 10–20 episodes) reflects the asynchronous protocol allowing tactical policies to stabilize before strategic updates commence. Low variance across runs confirms algorithmic stability.

C. Qualitative Analysis

Figure 4 shows the execution trajectory of a trained AHPG model over a complete episode. The strategic layer learns to assign non-overlapping regions that collectively tile the grid, initially placing agents in corners and then adapting assignments to unexplored areas. Agents balance region compliance with opportunistic resource collection. By frame (e), full resource collection with maintained spatial coordination validates the effectiveness of bidirectional hierarchical control.

VII. CONCLUSION

This paper introduced CHMAS, a hierarchical MARL framework based on bidirectional coupling between strategic and tactical layers. By enabling strategic planning with exclusive global state access while maintaining distributed tactical execution, CHMAS provides a new architectural paradigm for coordinating agents across multiple temporal

scales. The asynchronous update protocol, with a decaying strategic step size schedule, manages the non-stationarity inherent in hierarchical learning and admits $\mathcal{O}(\log K / \sqrt{K})$ convergence guarantees for the strategic layer.

The bidirectional coupling mechanism offers a principled approach to constrained reinforcement learning: strategic guidance encodes global constraints while tactical feedback ensures feasibility, with λ providing tunable constraint satisfaction.

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智能体 i 只观察到包含其位置、附近资源和战略指导的局部信息 $o_i^{\text{tac}} \in \mathbb{R}^{20}$ 。

策略性动作在每一轮开始时（每个 epoch 共 5 个时间步）为每个智能体分配一个 9×9 区域中心 $a_i^{\text{str}} = (g_i^x, g_i^y) \in [4, 20]^2$ 。战术智能体选择离散动作 $a_i^{\text{tac}} \in \{0, 1, 2, 3, 4, 5\}$ （移动、收集、等待）。

奖励结构实现了双向耦合：战术智能体在区域外获得 $r_{i,t}^{\text{tac}} = r_{i,t}^{\text{collect}} - 0.01 - 0.01 \cdot \mathbf{1}[\text{outside region}]$ ，而策略层则仅获得全局覆盖奖励 $R_k^{\text{global}} = 0.01 \times |\text{新访问的单元格}|$ 的奖励，该奖励由 s^{env} 计算得出。策略性奖励整合了多个方面： $R_k^{\text{str}} = R_k^{\text{global}} + \lambda \sum_{t,i} r_{i,t}^{\text{tac}}$ 以及 $\lambda = 0.5$ 。

该实现代码已公开发布在 https://github.com/EricDmWang/Hierarchy_RL.git。

B. 结果与分析

我们使用参数共享、Dueling 架构、双 Q 学习和优先经验回放的 Multi-Agent DQN (MA-DQN) 实现战术层。战略层采用全局状态上的集中式 DQN。训练运行了 400 个回合，使用 5 个独立的种子。

图 2 显示了战术层的学习进程。Q 值 (a) 在 300 个回合附近稳定前迅速增长。回报 (b) 从大约 -80 改善到 -10 ，显示智能体在学习高效资源收集的同时尊重区域约束。评价者损失 (c) 和策略损失 (d) 在探索期间早期达到峰值，然后随着策略稳定而下降。

图 3 显示了战略层的异步学习，每 $N_f = 5$ 个回合更新一次。战略 Q 值 (a) 从 -8 改善到接近 0，战略回报 (b) 从 -10 增加到大约 15，展示了全局覆盖目标的成功优化。学习延迟开始（前 10–20 个回合）反映了异步协议允许战术策略稳定后才开始战略更新。运行间方差低证实了算法稳定性。

C. 定性分析

图4展示了训练好的AHPG模型在一个完整回合中的执行轨迹。策略层学习分配不重叠的区域，这些区域共同铺满网格，最初将智能体放置在角落，然后调整分配以探索未知的区域。智能体在区域合规性和机会性资源收集之间取得平衡。到帧(e)时，完全的资源收集与保持空间协调验证了双向分层控制的有效性。

VII. 结论

本文介绍了CHMAS，一个基于战略层和战术层之间双向耦合的分层多智能体强化学习框架。通过实现具有独占全局状态访问的战略规划，同时保持分布式战术执行，CHMAS为跨多个时间尺度的智能体协调提供了一种新的架构范式。

异步更新协议，结合衰减的战略步长调度，管理了分层学习中固有的非平稳性，并允许 $\mathcal{O}(\log K / \sqrt{K})$ 收敛保证用于战略层。

双向耦合机制为约束强化学习提供了一种原则性方法：策略指导编码全局约束，而战术反馈确保可行性， λ 提供可调的约束满足。

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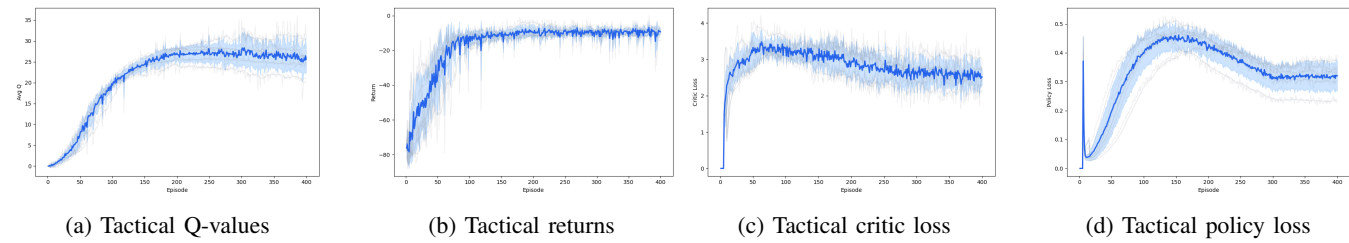


Fig. 2: Tactical layer learning dynamics.

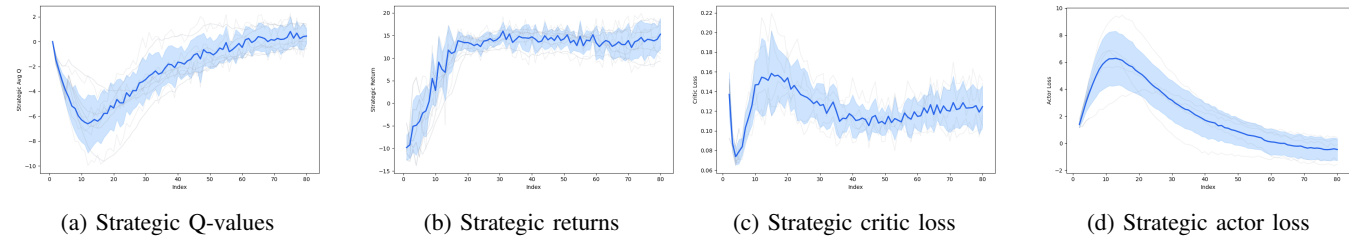


Fig. 3: Strategic layer learning dynamics with $N_f = 5$.

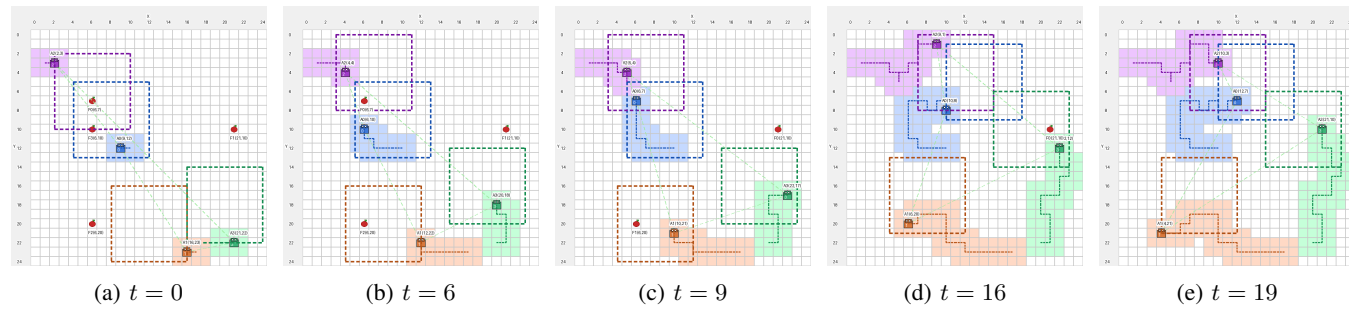


Fig. 4: Evolution of hierarchical coordination during execution. Agents operate within their assigned 9×9 regions (dashed boxes) while collectively achieving grid coverage. Red apples represent resources collected opportunistically.

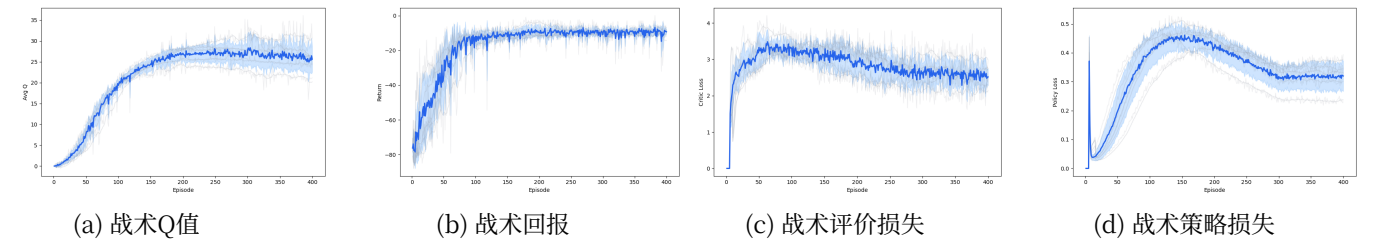


图2: 战术层学习动态。

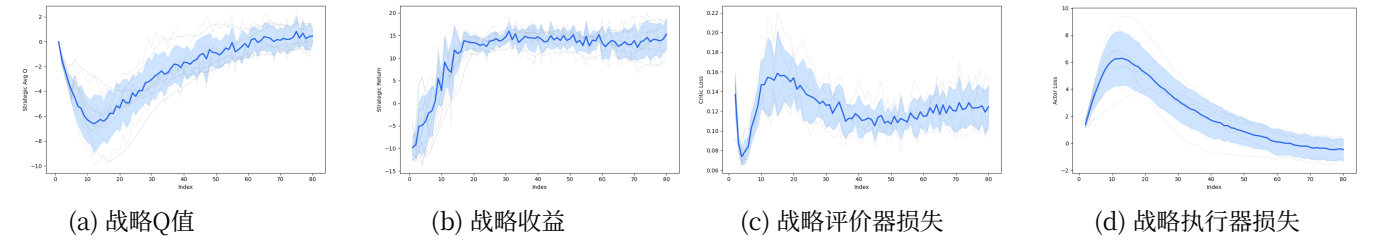


图3: 具有 $N_f = 5$ 的战略层学习动态

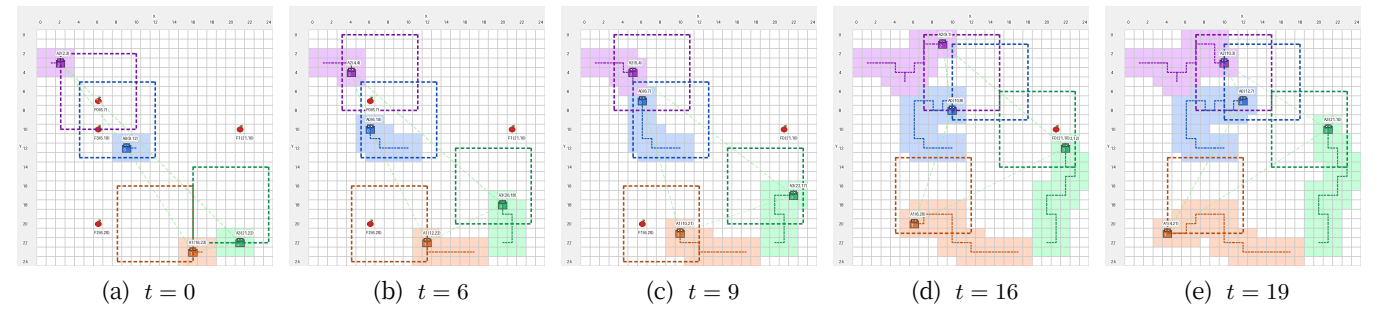


图4: 执行过程中分层协调的演化。智能体在其分配的 9×9 区域内（虚线框）运行，同时共同实现网格覆盖。红色苹果代表机会性收集的资源。

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